

Chaos Canvas of Lorentz Attractor

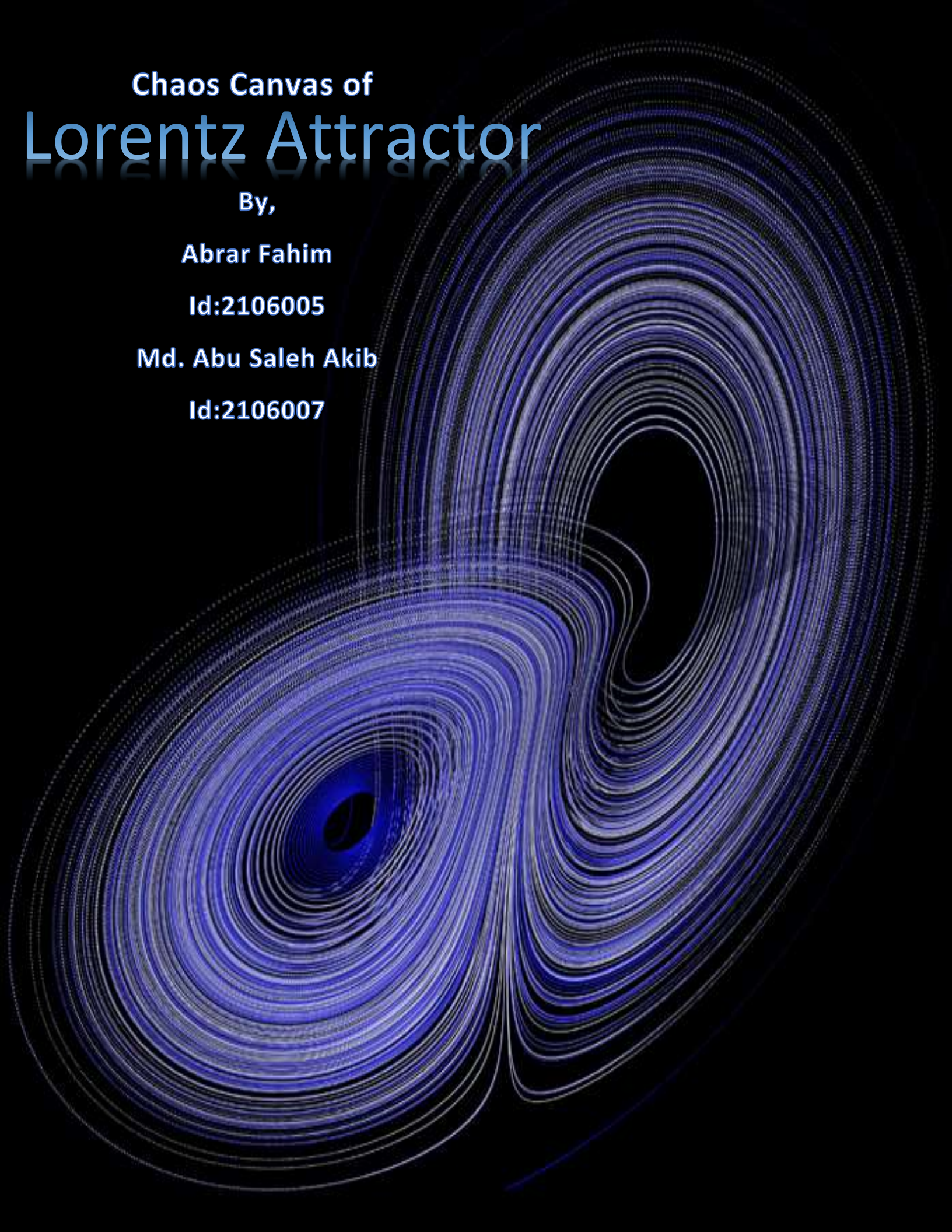
By,

Abrar Fahim

Id:2106005

Md. Abu Saleh Akib

Id:2106007



Project Title: Numerical Solution of Lorentz Attractor

Problem Statement: Solve Lorenz Attractor Equation, i.e. a first order simultaneous equation, numerically and plot the characteristics of the equation.

Objectives:

1. To take system parameters (Sigma, Rho and Beta).
2. To solve the system by Runge-Kutta method.
3. To display time plot and phase plot and a 3D plot as well.

Introduction

Lorenz attractor system is a set of linear equations modeled by Edward Lorenz to determine weather forecast. The system is both chaotic and deterministic and very sensitive to initial conditions.

$$\frac{dx}{dt} = \sigma(y - x),$$

$$\frac{dy}{dt} = x(\rho - z) - y$$

$$\frac{dz}{dt} = xy - \beta z.$$

The system is dependent on three parameters known as σ (sigma), β (beta) and ρ (rho).

If $\rho < 1$, the system is called 'Global Attractor' and its behavior is not chaotic.

If $\rho > 1$, the system can display two kinds of nature depending on the parameters.

1. Chaotic

2. Damping

1. Chaotic:

Two conditions must satisfy to bring the system in chaotic nature, also known as "Pitchfork Bifurcation". The conditions are :

$$\rho < \sigma \frac{\sigma + \beta + 3}{\sigma - \beta - 1},$$
$$\sigma > \beta + 1$$

Chaotic system is both deterministic and unpredictable as for certain values of parameters specific output will be found but it can never be known what the next output can be. It is because the slightest change in the initial values can cause drastic change in chaotic system.

It should be noted that the chaotic system generates a butterfly like phase plot, which later revolutionized the idea of "The Butterfly Effect" .

2. Damping:

If the given conditions are not fulfilled but $\rho > 1$, then it is called a damping system. The phase plot this time is a beautiful spiral shape. Also damping system is not as unstable as chaotic system, so significant change of initial values have small effect on the final output.

Procedure

A MATLAB app was built to conduct and display the plot of Lorenz Attractor under different conditions.

At first, the inputs were to be taken from the users. So dialog boxes were created to receive the values of parameters and initial values from the users using the code below. Along with the initial value, the functions for the 'Runge-Kutta' method was also given here :

```

sigma=app.SigmaEditField.Value;
beta=app.BetaEditField.Value;
rho=app.RhoEditField.Value;

f = @(t, x, y, z) sigma * (y - x);
g = @(t, x, y, z) x * (rho - z) - y;
r = @(t, x, y, z) x * y - beta * z;

t0=0;
x0=app.X0EditField.Value;
y0=app.Y0EditField.Value;
z0=app.Z0EditField.Value;
tn=app.FinalTimeEditField.Value;
h=app.StepSizeEditField.Value;
n=(tn-t0)/h;
t(1)=t0;
x(1)=x0;
y(1)=y0;
z(1)=z0;

```

Once the parameters and the initial values are taken, it is time to solve the differential equation using “Ruhge-Kutta’ method. The code for this purpose is given below :


```

% Runge-Kutta method
for i = 1:n
    t(i+1) = t0 + i*h;

    k1 = h*f(t(i), x(i), y(i), z(i));
    m1 = h*g(t(i), x(i), y(i), z(i));
    l1 = h*r(t(i), x(i), y(i), z(i));

    k2 = h*f(t(i) + h/2, x(i) + k1/2, y(i) + m1/2, z(i) + l1/2);
    m2 = h*g(t(i) + h/2, x(i) + k1/2, y(i) + m1/2, z(i) + l1/2);
    l2 = h*r(t(i) + h/2, x(i) + k1/2, y(i) + m1/2, z(i) + l1/2);

    k3 = h*f(t(i) + h/2, x(i) + k2/2, y(i) + m2/2, z(i) + l2/2);
    m3 = h*g(t(i) + h/2, x(i) + k2/2, y(i) + m2/2, z(i) + l2/2);
    l3 = h*r(t(i) + h/2, x(i) + k2/2, y(i) + m2/2, z(i) + l2/2);

    k4 = h*f(t(i) + h, x(i) + k3, y(i) + m3, z(i) + l3);
    m4 = h*g(t(i) + h, x(i) + k3, y(i) + m3, z(i) + l3);
    l4 = h*r(t(i) + h, x(i) + k3, y(i) + m3, z(i) + l3);

    x(i+1) = x(i) + (1/6)*(k1 + 2*k2 + 2*k3 + k4);
    y(i+1) = y(i) + (1/6)*(m1 + 2*m2 + 2*m3 + m4);
    z(i+1) = z(i) + (1/6)*(l1 + 2*l2 + 2*l3 + l4);

```

Now that the differential equations are solved, it is time to observe the plot. A switch case has been used here to select and observe the plot user wishes to observe. The code is given below :

```

switch app.PlottypeDropDown.Value
case 'x vs t'
    xlabel(app.UIAxes, 't');
    ylabel(app.UIAxes, 'x');
    title(app.UIAxes, 'x vs t');
    plot(app.UIAxes, t, x);
    view(app.UIAxes, 0, 90);
case 'y vs t'
    xlabel(app.UIAxes, 't');
    ylabel(app.UIAxes, 'y');
    title(app.UIAxes, 'y vs t');
    plot(app.UIAxes, t, y);
    view(app.UIAxes, 0, 90);
case 'z vs t'
    xlabel(app.UIAxes, 't');
    ylabel(app.UIAxes, 'z');
    title(app.UIAxes, 'z vs t');
    plot(app.UIAxes, t, z);
    view(app.UIAxes, 0, 90);

case 'z vs x'
    xlabel(app.UIAxes, 'x');
    ylabel(app.UIAxes, 'z');
    title(app.UIAxes, 'z vs x');
    plot(app.UIAxes, x, z, 'green');
    view(app.UIAxes, 0, 90);

case 'y vs x'
    xlabel(app.UIAxes, 'x');
    ylabel(app.UIAxes, 'y');
    title(app.UIAxes, 'y vs x');
    plot(app.UIAxes, x, y, 'red');
    view(app.UIAxes, 0, 90);

case 'z vs y vs x'
    xlabel(app.UIAxes, 'x');
    ylabel(app.UIAxes, 'y');
    zlabel(app.UIAxes, 'z');
    title(app.UIAxes, 'z vs y vs x');
    plot3(app.UIAxes, x, y, z);
end

```

This is how an user can observe the plot of solution of Lorenz Attractor. But the plot observed here shows the final output only. Another feature is also added here that can showcase the progression of the plots with respect to time. Thus an user can decide a time range within which he can observe the system and its progression.

Real Time Plot :

For real time plotting, the inputs and the “Runge-Kutta” solution was same as the static plot. But a little modification was done in the plot part. The modified code is given below :

```
% Plotting with reduced refresh rate
if mod(i, refreshRate) == 0 || i == n
    switch app.DOptionsDropDown.Value
        case 'x vs t'
            xlabel(app.UIAxes2, 't');
            ylabel(app.UIAxes2, 'x');
            plot(app.UIAxes2, t(1:i+1), x(1:i+1), 'red');
        case 'y vs t'
            xlabel(app.UIAxes2, 't');
            ylabel(app.UIAxes2, 'y');
            plot(app.UIAxes2, t(1:i+1), y(1:i+1), 'green');
        case 'z vs t'
            xlabel(app.UIAxes2, 't');
            ylabel(app.UIAxes2, 'z');
            plot(app.UIAxes2, t(1:i+1), z(1:i+1), 'blue');
        case 'y vs x'
            xlabel(app.UIAxes2, 'x');
            ylabel(app.UIAxes2, 'y');
            plot(app.UIAxes2, x(1:i+1), y(1:i+1), 'green');
        case 'z vs x'
            xlabel(app.UIAxes2, 'x');
            ylabel(app.UIAxes2, 'z');
            plot(app.UIAxes2, x(1:i+1), z(1:i+1), 'red');
    end
end
```



```

% Plot x vs y vs z
xlabel(app.UIAxes3,'x');
ylabel(app.UIAxes3,'y');
zlabel(app.UIAxes3,'z');

plot3(app.UIAxes3,x(1:i+1), y(1:i+1), z(1:i+1), 'b');
drawnow;

end
end

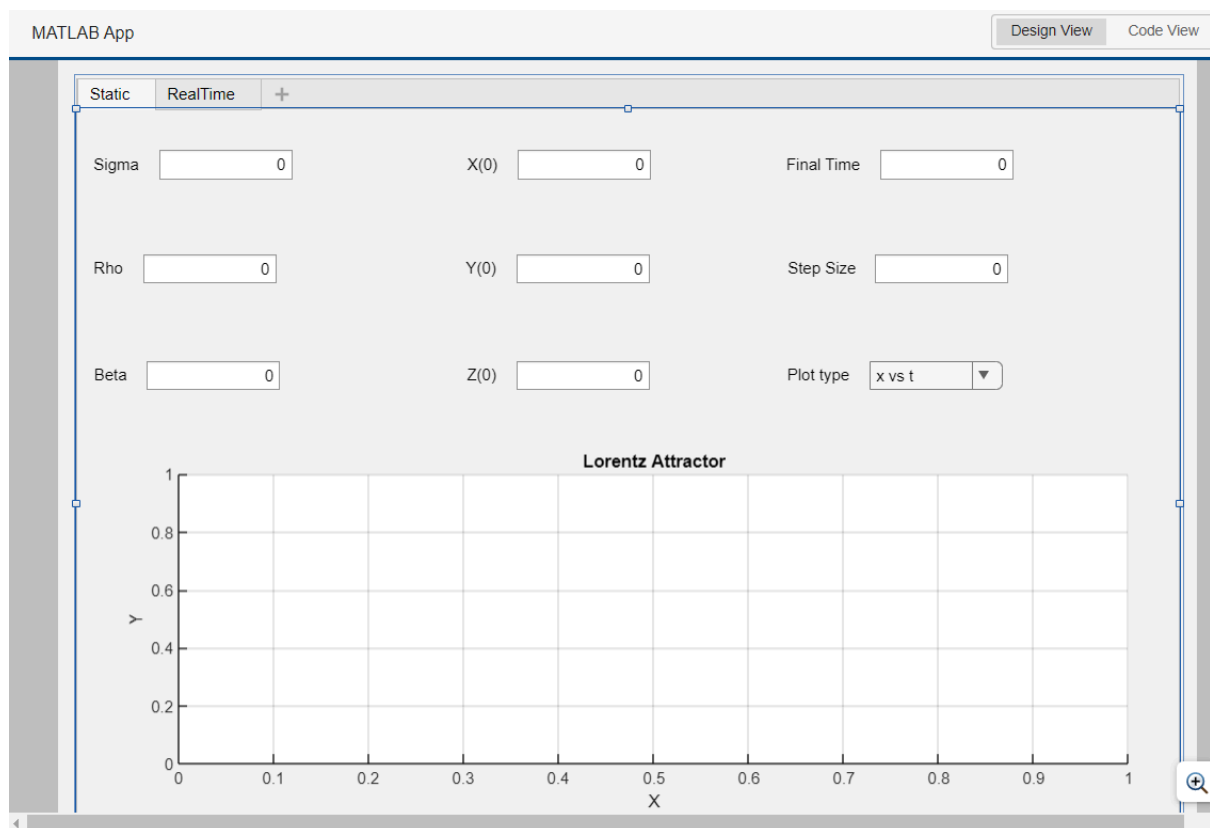
```

Input and Output

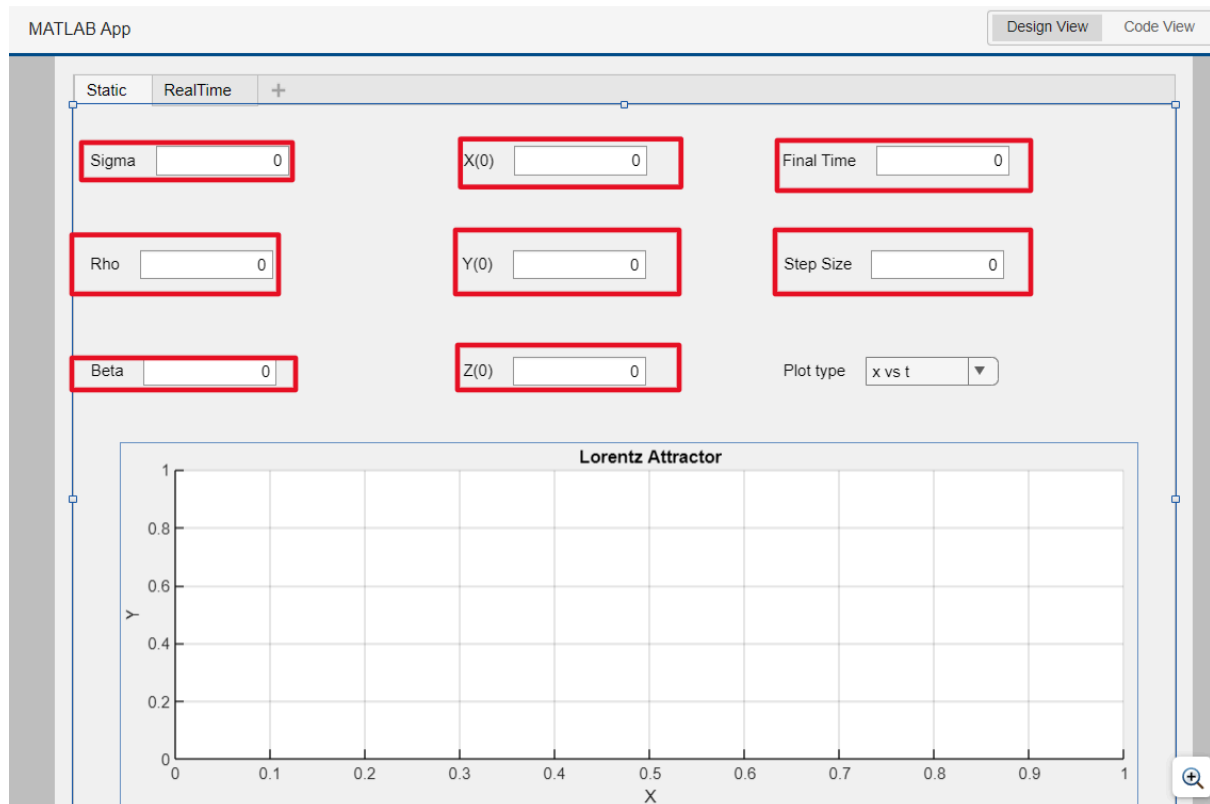
Static Plot :

Input :

Upon opening the “Lorenz_Attractor.mlapp” file, the following app will be visible on the screen for the user.



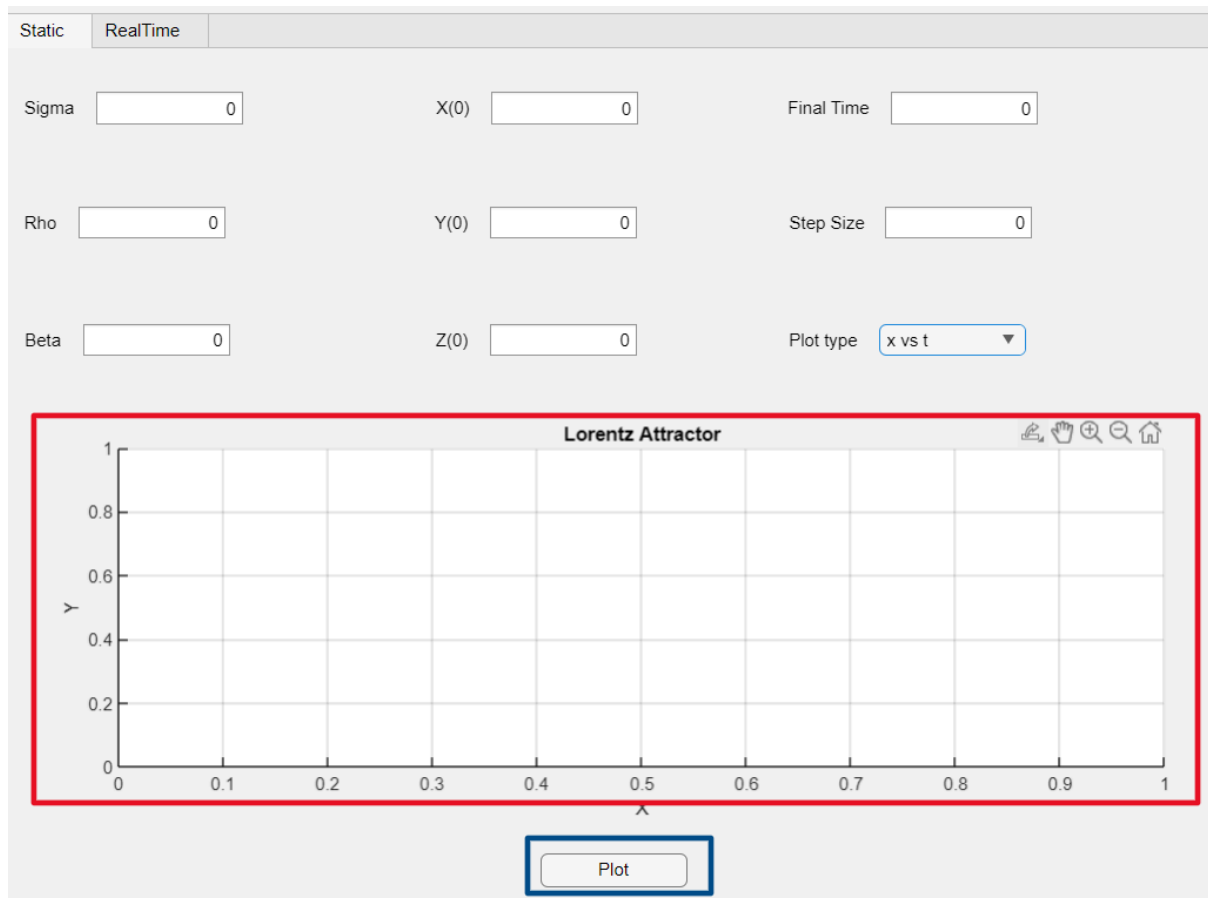
Here the user at first has to put value in the following boxes :



Then the user must select the type of plot to be seen from the dialog box of plot type.

Output:

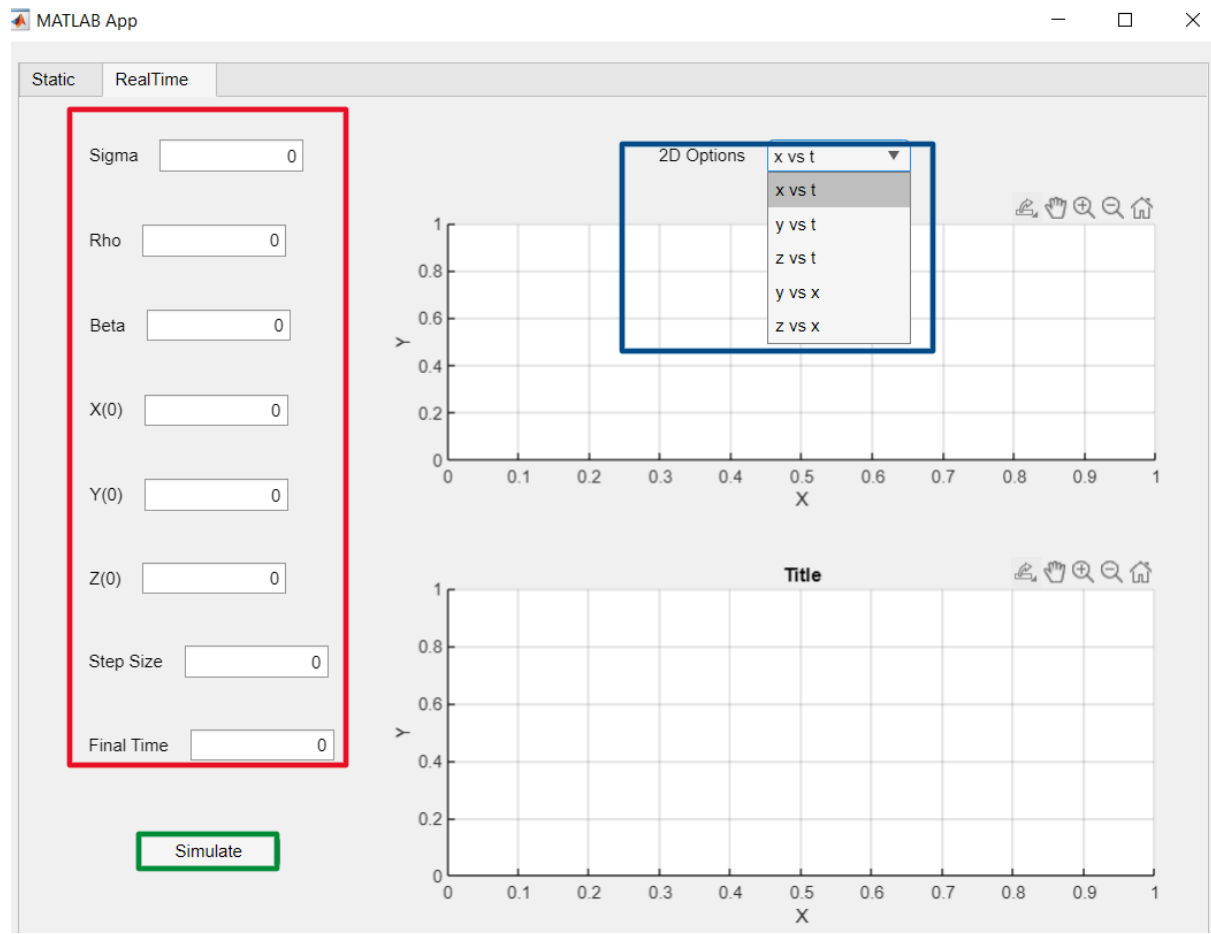
After selecting the type of plot, the plot button needs to be clicked and the plot will be visible in the menu below :



Real Time Plot:

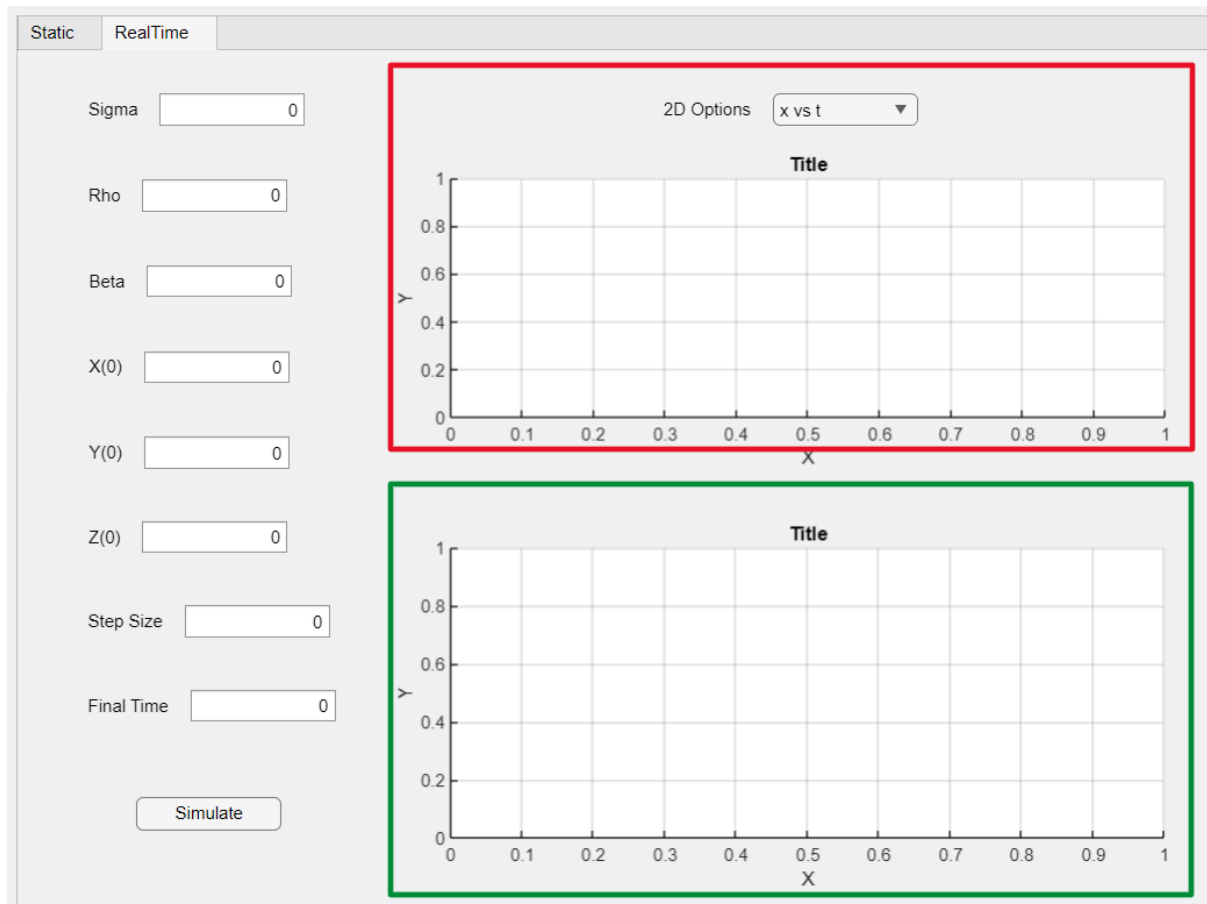
Input :

The input for real time plotting is almost same as the static plot. At first the user must give input for the parameters and initial values and press SIMULATE button. There is also a dialog box for 2D plot type.



Output :

There are two windows for output here. One is showing the selected 2D plot. And the other one is showing overall 3D plot of the system.



Sample Results

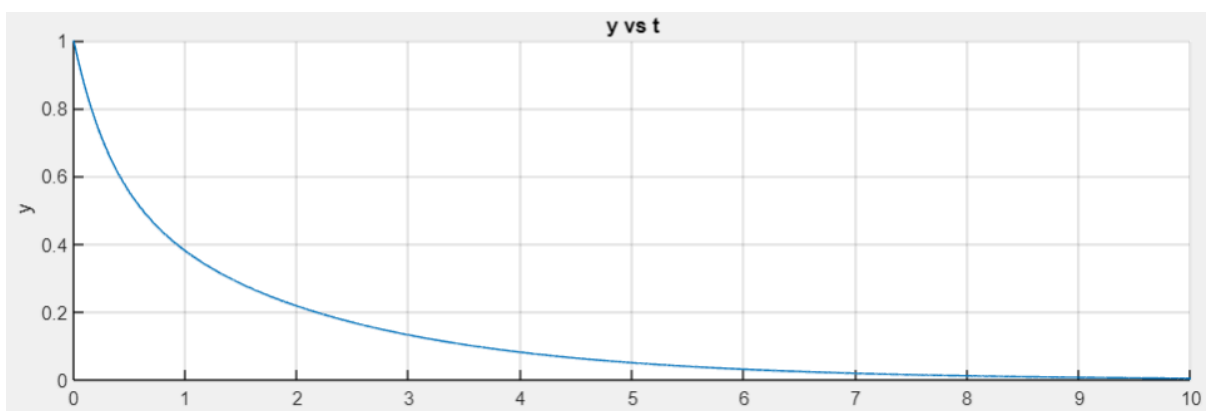
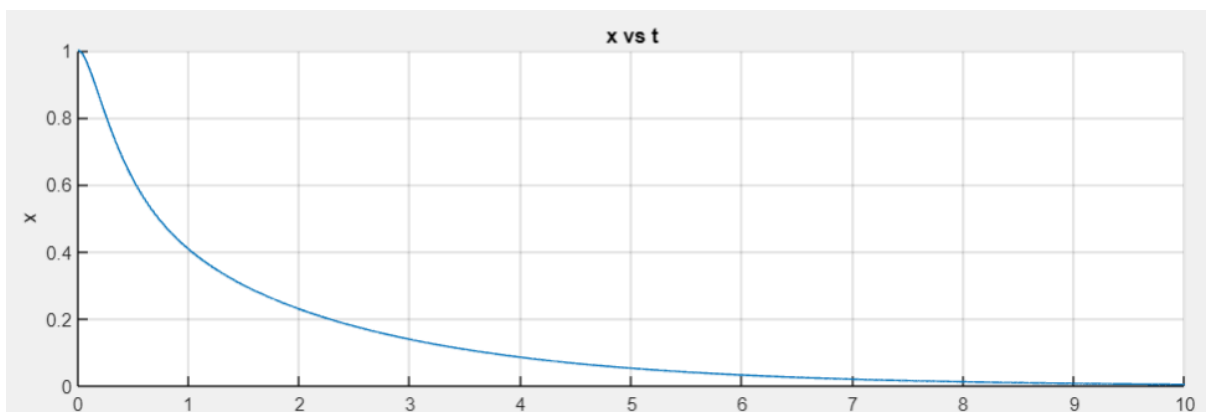
For sample results we would like to use the same values for parameters and initial values used in the main experiment. So initial values will be 1 for x,y and z and $\sigma(\text{sigma})=10$ and $\beta(\text{beta})=8/3$. We will change the value of $\rho(\text{rho})$ to observe the change in systems.

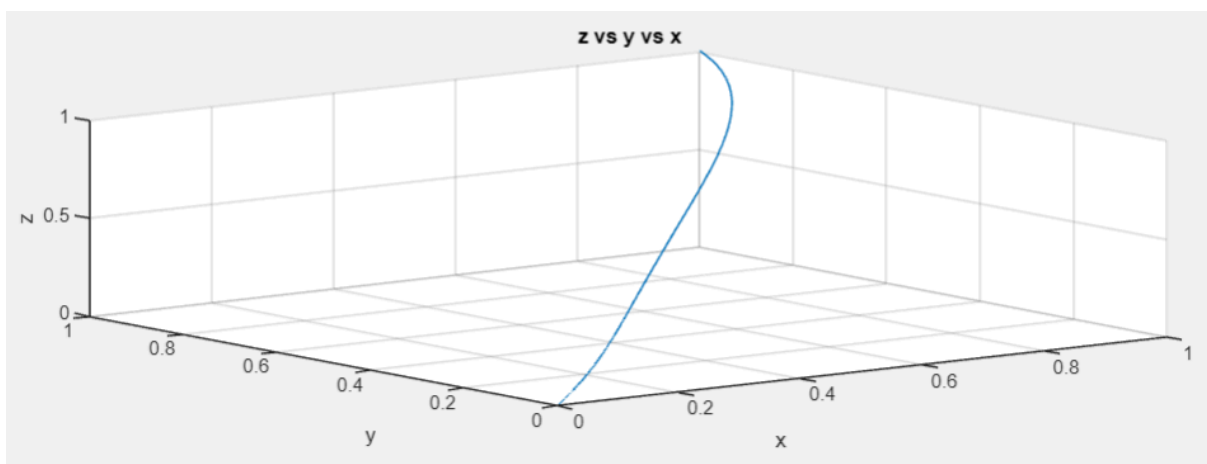
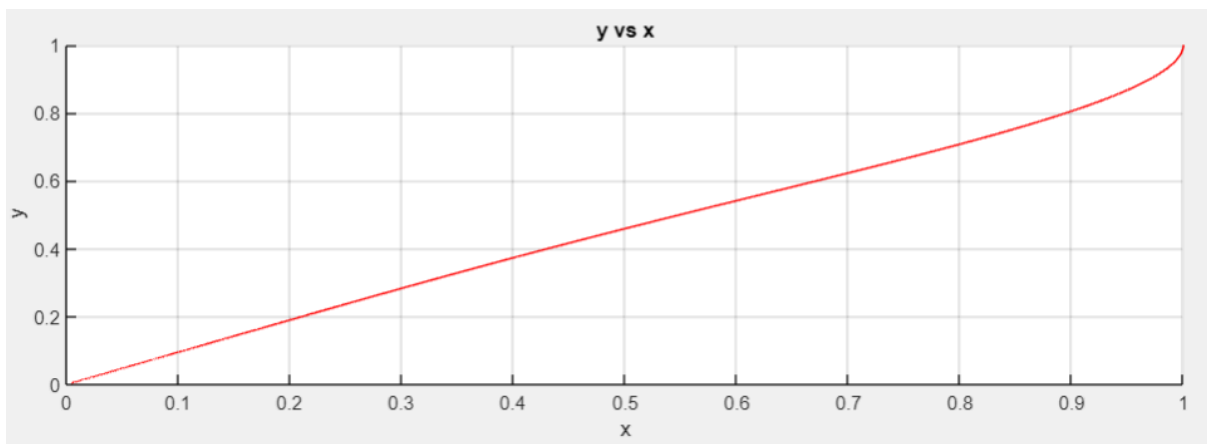
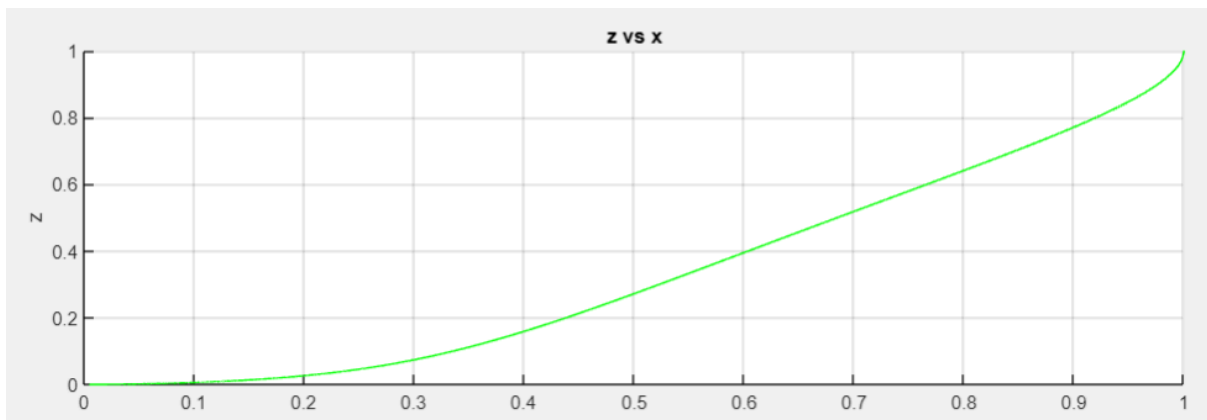
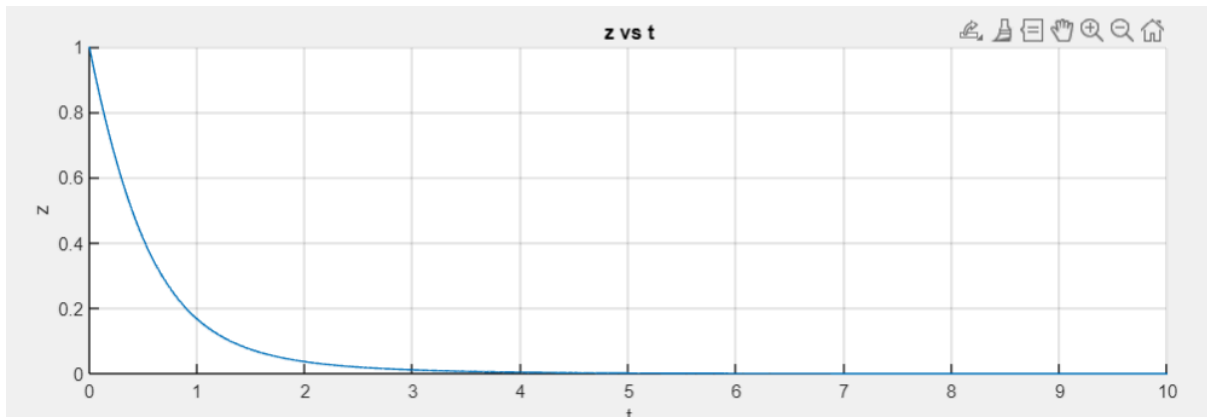
1. Global Attractor ($\rho < 1$) :

Input :

Sigma	10	X(0)	1	Final Time	10
Rho	0.5	Y(0)	1	Step Size	0.0001
Beta	2.67	Z(0)	1	Plot type	x vs t

Output :





In this system, the output is converging to zero (0) with time.

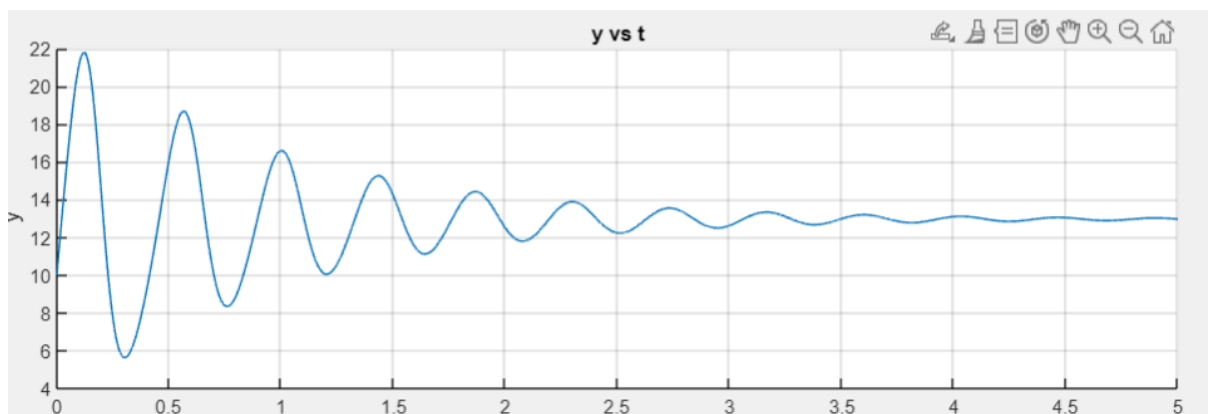
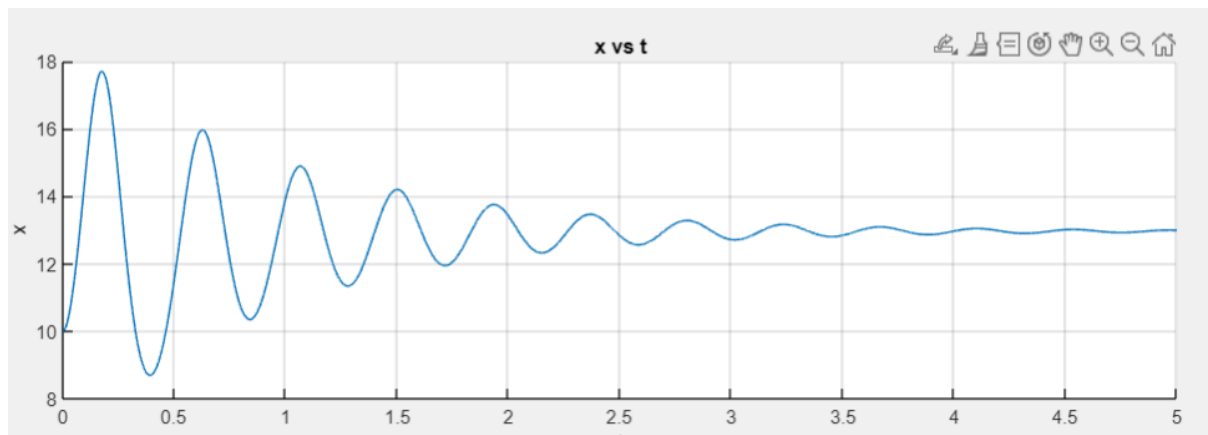
2. Damping System :

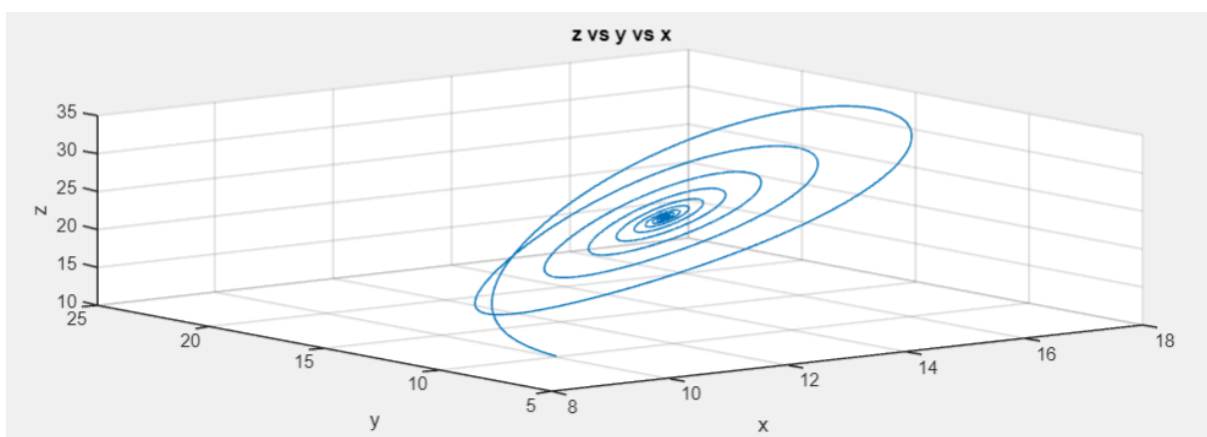
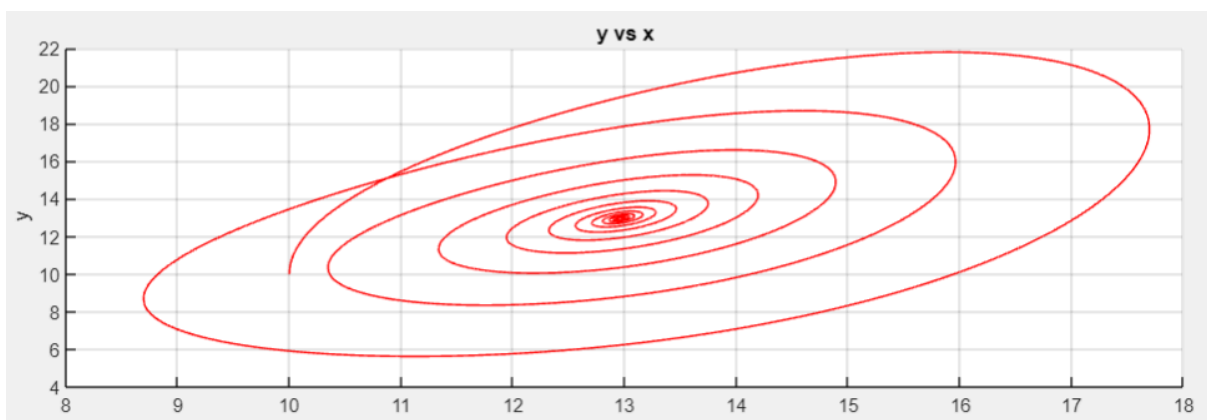
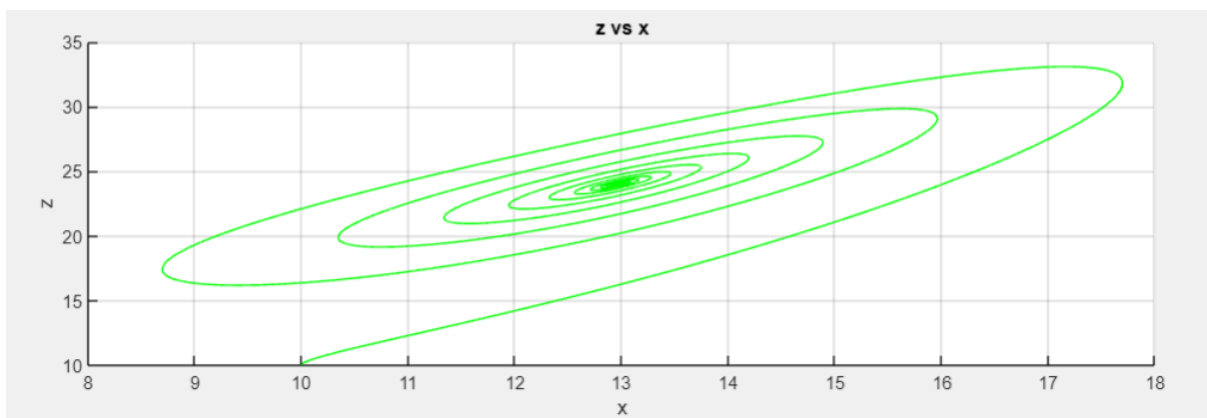
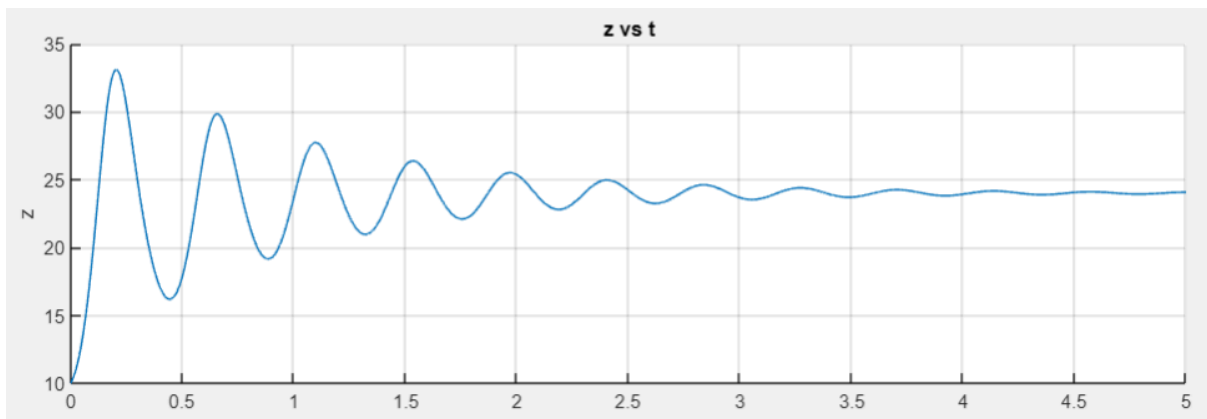
To get the system act as a damping system, the parameters and initial values were taken as seemed suitable.

Input :

Sigma	10	X(0)	10	Final Time	5
Rho	25	Y(0)	10	Step Size	0.0001
Beta	7	Z(0)	10	Plot type	z vs x

Output :



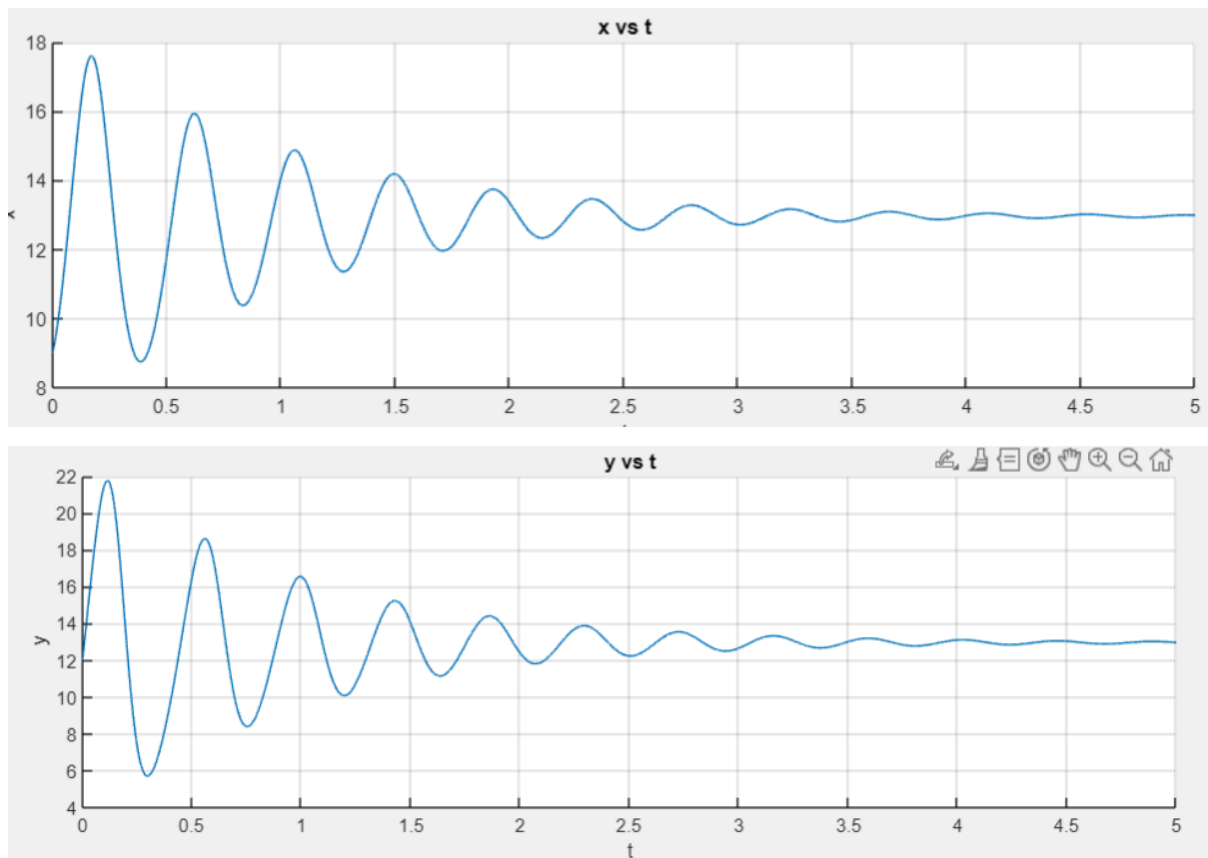


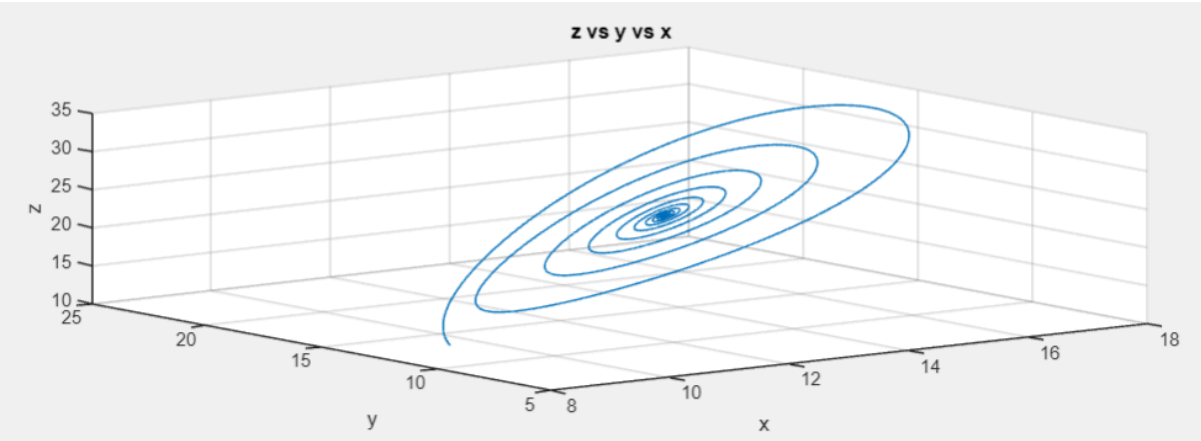
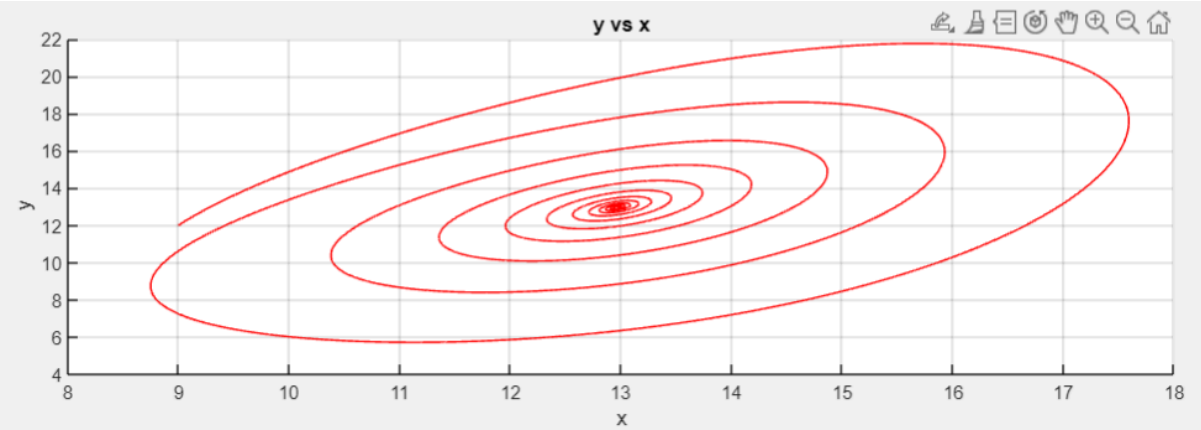
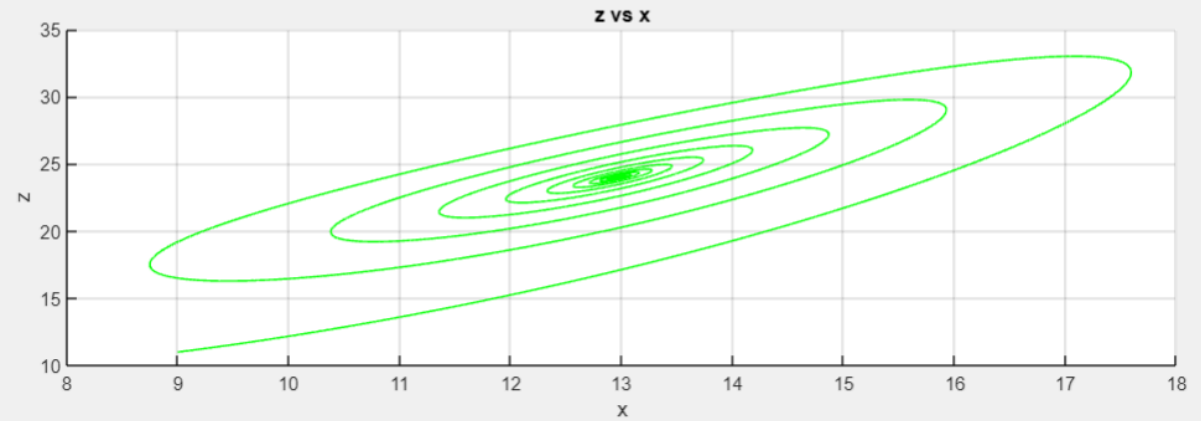
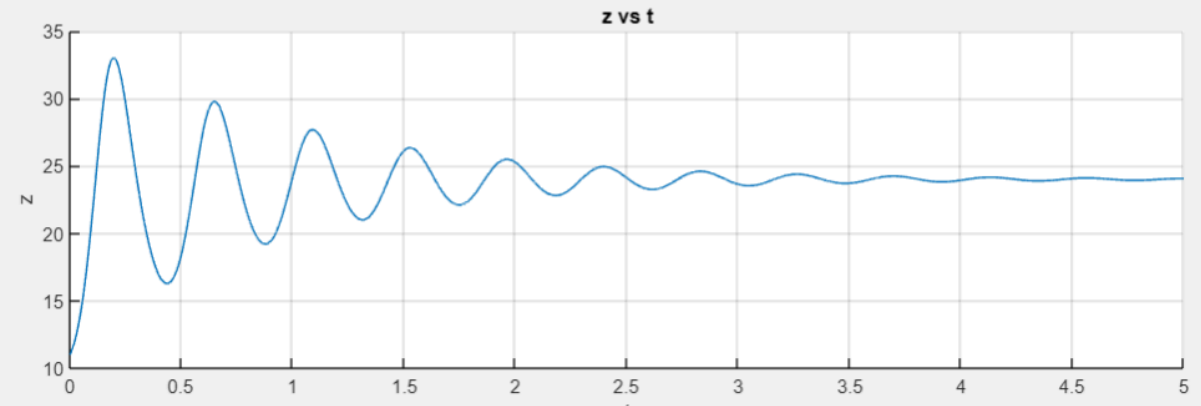
Here we can see the spiral phase plot as expected. Now let's change the initial values slightly and observe the plots again.

New Input :

Sigma	10	X(0)	9	Final Time	5
Rho	25	Y(0)	12	Step Size	0.0001
Beta	7	Z(0)	11	Plot type	x vs t

Output :





Even after changing the initial values, no significant change is observed as it should be.

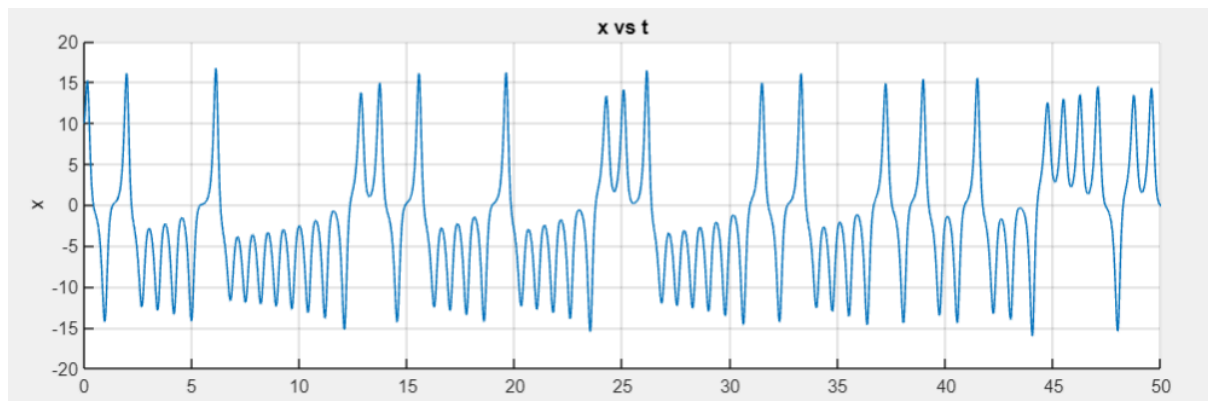
3. Chaotic System :

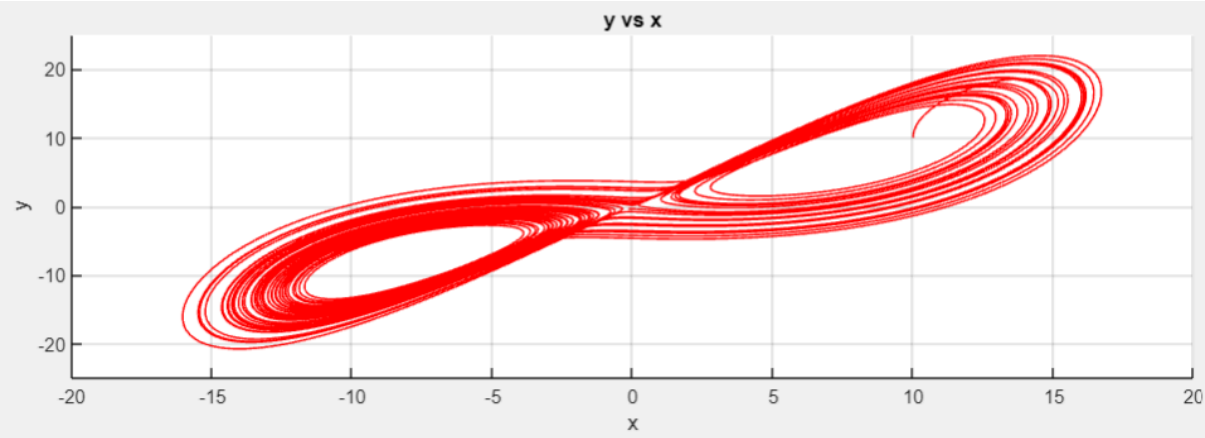
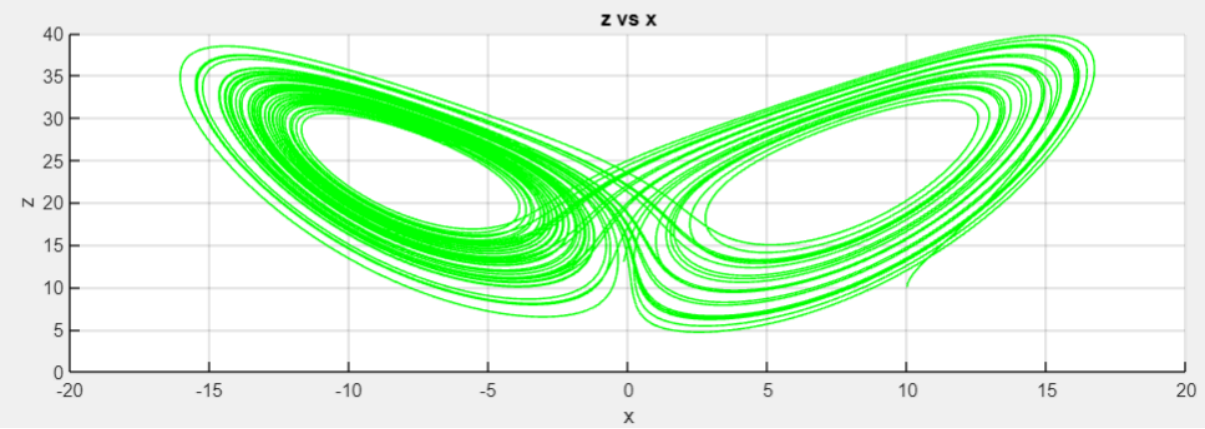
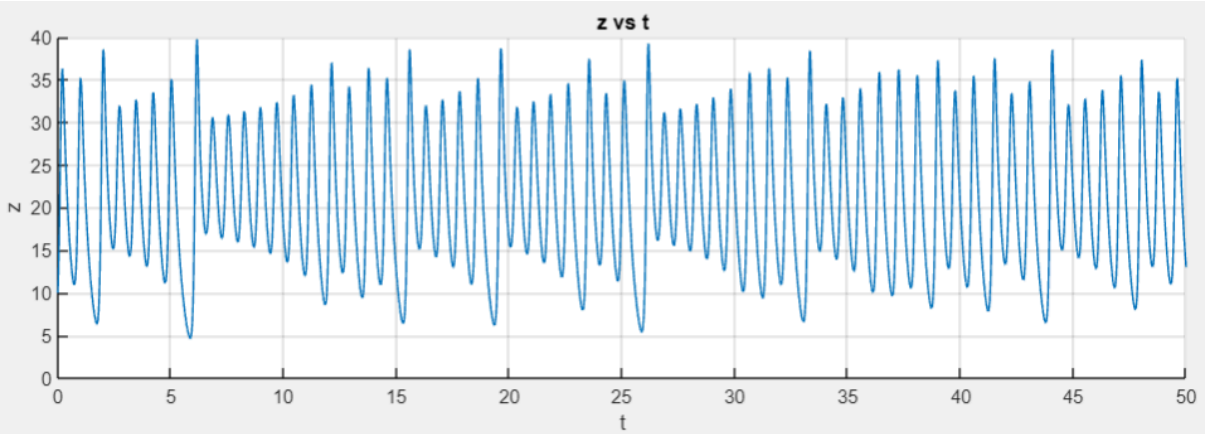
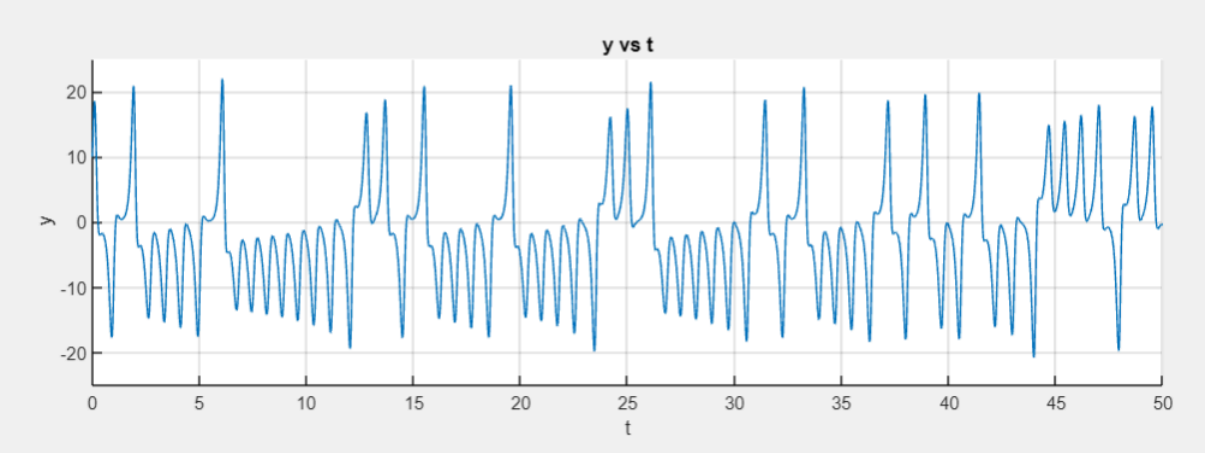
Input values were again adjusted to achieve a chaotic system.

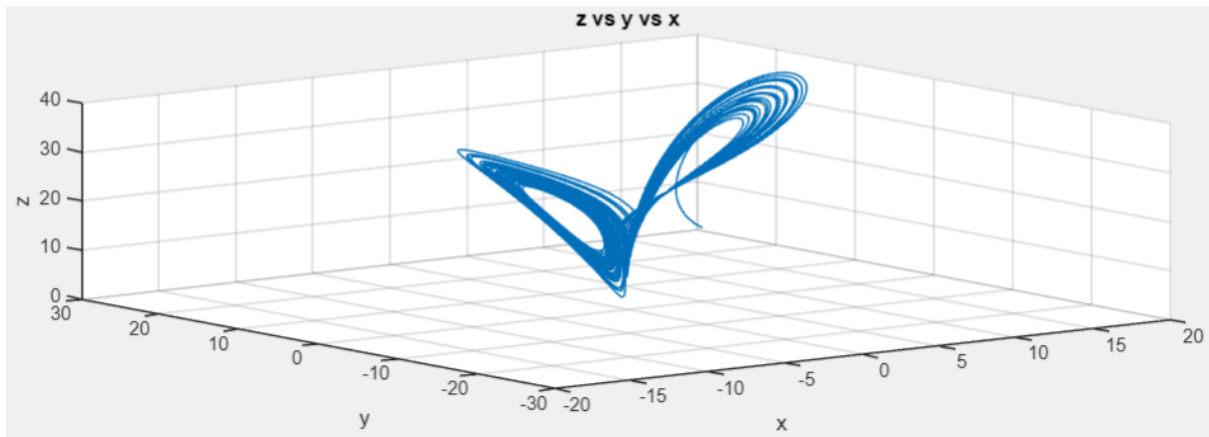
Input :

Sigma	10	X(0)	10	Final Time	50
Rho	25	Y(0)	10	Step Size	0.001
Beta	2.67	Z(0)	10	Plot type	x vs t

Output :





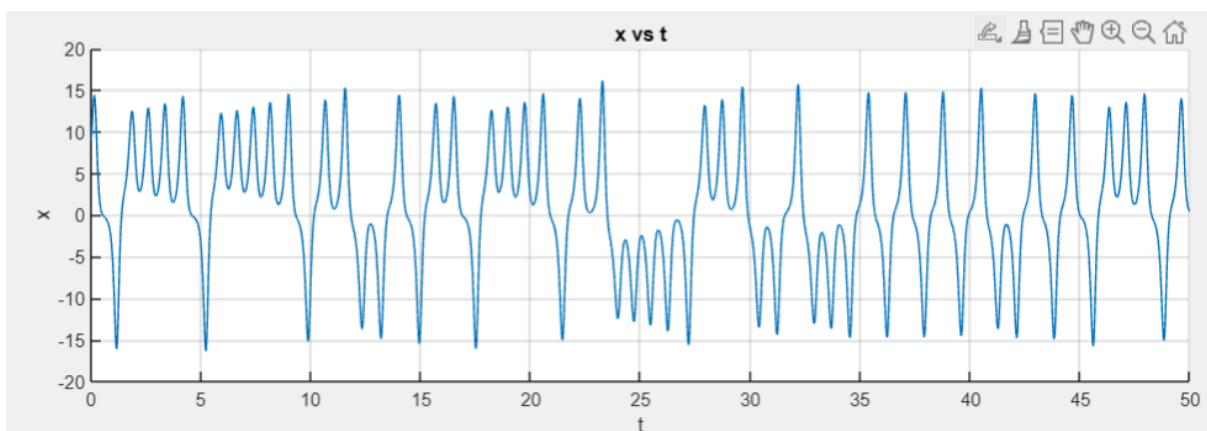


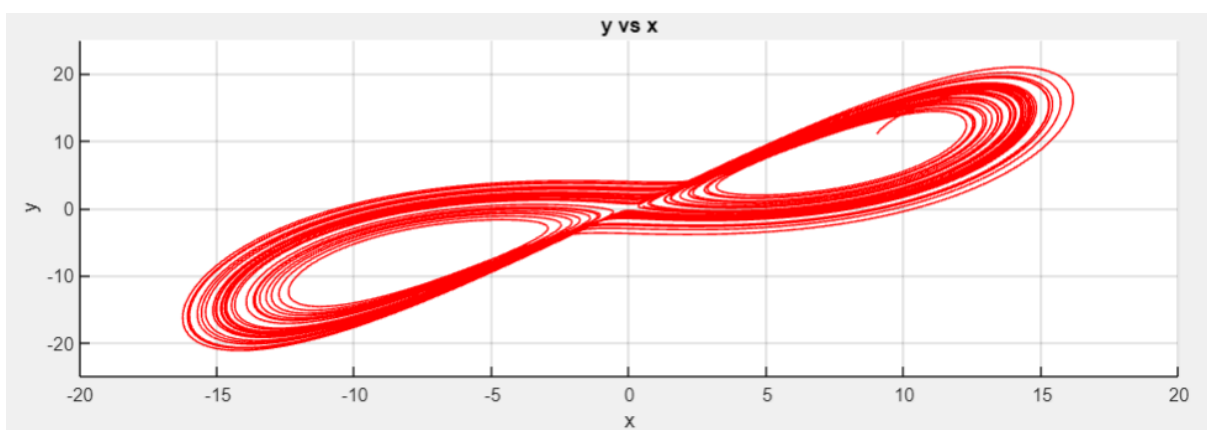
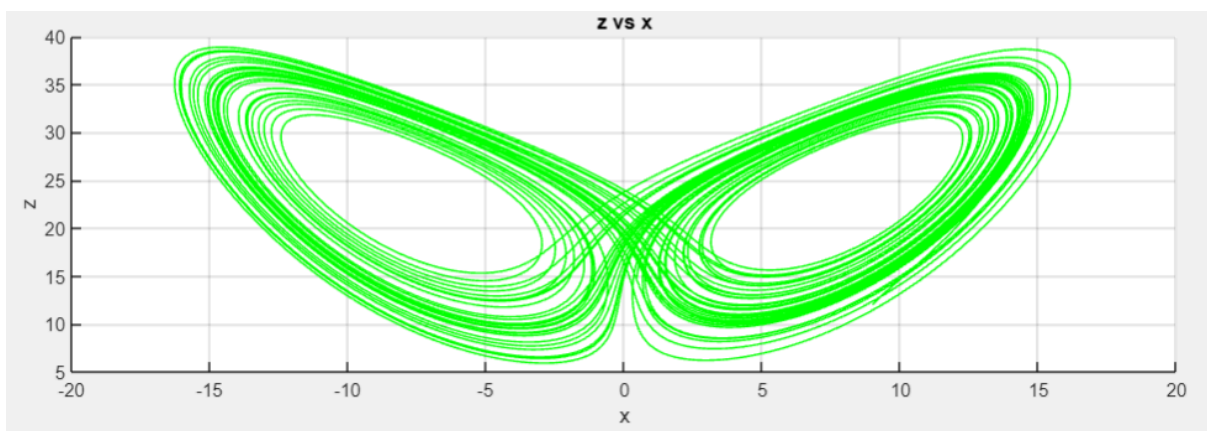
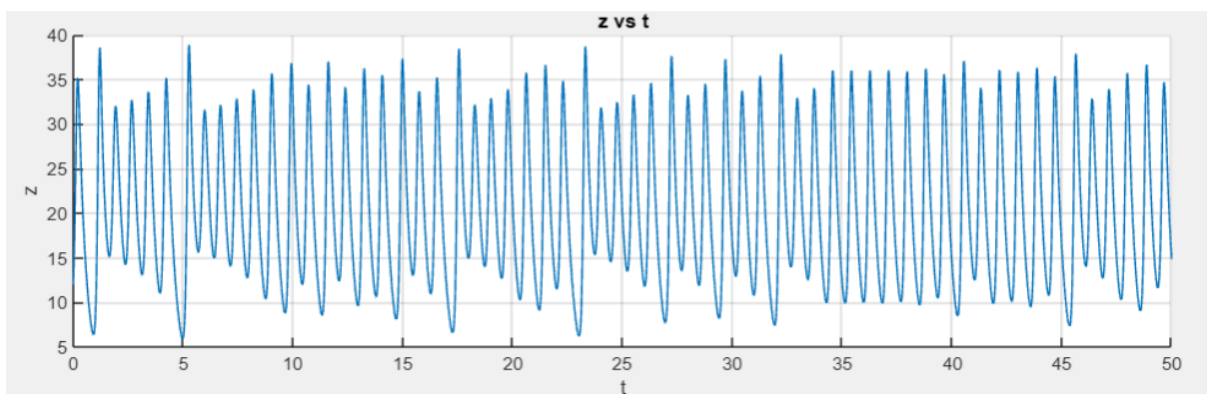
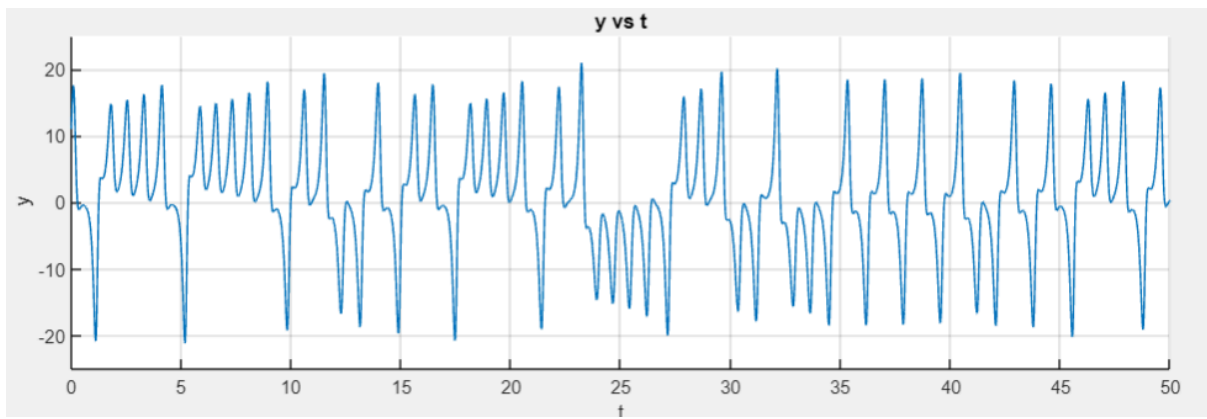
The phase plot in this system looks like a butterfly, as it should. Now let's change the initial values slightly and observe the new outputs.

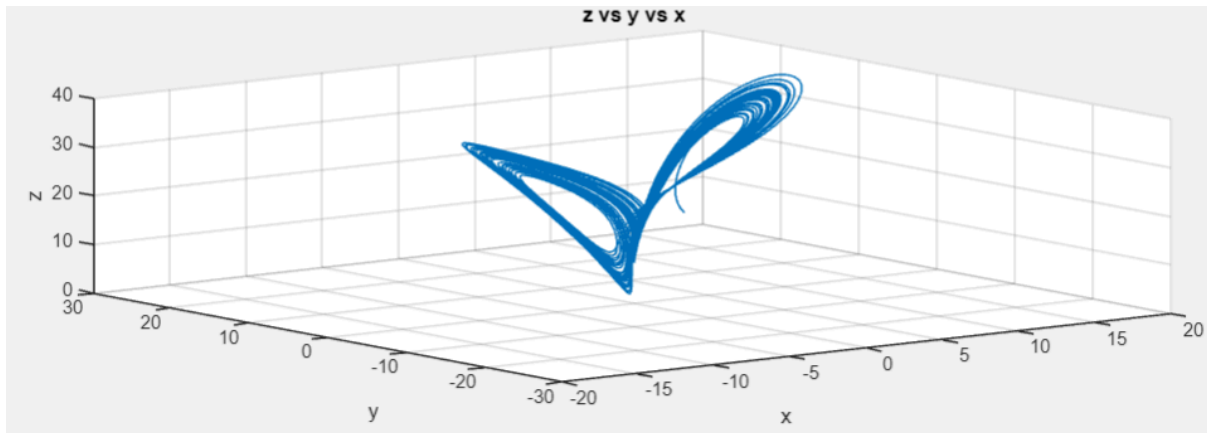
New Input :

Sigma	10	X(0)	9	Final Time	50
Rho	25	Y(0)	11	Step Size	0.001
Beta	2.67	Z(0)	12	Plot type	x vs t

Output :







Here, slight change in input values are causing significant change in the plots, which also support the theory.

Conclusion

The purpose of Lorenz Attractor system is to decide the characteristic of a system by solving series of differential equations. According to the experiment by Lorenz, a system may be damping where the system converges to a certain value and becomes stable, or chaotic where the output of the system is completely unpredictable, or it may be a global attractor where all the values eventually become zero (0). Now the characteristic a system shows depends on the values of the parameters (σ , β , ρ) of the equations.

So, for desired values of the parameters, the x vs t , y vs t , z vs t , y vs x , z vs x and a 3D plot (x vs y vs z) is also shown. Apart from the final output an extra option is also available to watch the real time plot. It means the plot will be made live and a better idea can be gotten of how the final plot is generated.

Limitations

The system shows the desired outputs for values of parameters $\sigma, \beta, \rho > 0$.

If $\sigma, \beta, \rho = 0$ but the initial conditions are non-zero, then the value of x is constant all the time but y and z shows damping value and eventually converge to a constant value as well.

If $\sigma, \beta, \rho = 0$ and all the initial values are also zero (0), then all x, y and z show constant output.

If $\sigma, \beta, \rho < 0$, that means if the parameters are negative, the system shows bizarre output which is not defined in the Lorenz system.